

The two-row $d=4$ fiber law is an *imaginary-part* non-vanishing: a robust reduction and the boundary families

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Abstract

For $\lambda = (a, b)$ with $a \geq b \geq 0$ and $a + b = 2m$, put $G_{(a,b)} = \langle s_{(a,b)}, \psi^m \rangle$ with $\psi = h_2 + i e_2$, $i = \sqrt{-1}$, a Gaussian integer. The two-row $d=4$ conjecture is $G_{(a,b)} = 0 \iff (a, b) = (2, 2)$. We give a new reduction that replaces the previously studied marginal continued-fraction envelope by a *robust, discrete* statement: writing b for the second row,

$$G_{(2m-b,b)} = [u^b]((1-u)(1+su+u^2)^m), \quad s = 1+i,$$

and

$$\operatorname{Im} G_{(2m-b,b)} = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} (-1)^{(k-1)/2} \binom{m}{k} [T(m-k, b-k) - T(m-k, b-k-1)],$$

where $T(N, j) = [u^j](1+u+u^2)^N$ are the (non-negative, unimodal) trinomial coefficients. Each bracket is ≥ 0 , so $\operatorname{Im} G$ is an alternating sum of non-negative integers. The law follows from the single statement

$$\operatorname{Im} G_{(2m-b,b)} \neq 0 \quad \text{for all } 1 \leq b \leq m, (m, b) \neq (2, 2).$$

We prove this in closed form for the four boundary families $b = 1, 2, 3, 4$ (in particular $G_{(2m-1,1)} = (m-1) + m i$ and $G_{(2m-2,2)} = m(m-2) i$, the latter being the *only* vanisher), we show the global minimum $\min_i |\operatorname{Im} G| = m$ is attained at the hook $b = 1$, and we verify $|\operatorname{Im} G_{(2m-b,b)}| \geq m$ for all $m \leq 300$. The point of the reduction is that the binding margin, which looked like a knife-edge $\frac{1}{4}$ in continued-fraction coordinates, is in truth an unbounded gap $\geq m$; the marginality was a normalisation artifact. The remaining gap is precisely the uniform non-vanishing of the displayed alternating trinomial sum for $b \geq 5$, and we record two concrete routes to it.

1 The fiber value as a single coefficient

Throughout $s = 1 + i$, so $s^2 = 2i$ and $\bar{s} = 1 - i$. For a two-row shape $\lambda = (a, b)$, $a + b = 2m$, $a \geq b \geq 0$, the fiber value is the adjacent difference of coefficients of the self-reciprocal polynomial $(1 + su + u^2)^m$:

$$r_l := [u^l](1 + su + u^2)^m, \quad G_{(2m-b,b)} = r_b - r_{b-1}. \tag{1}$$

This is the standard “master identity” specialised to $d = 4$ and two rows (equivalently $(1+i)^m G_\lambda = \sum_k \binom{m}{k} i^k \chi^\lambda(2^{m-k} 1^{2k})$). Since $r_b - r_{b-1} = [u^b]((1-u)(1+su+u^2)^m)$, we have

$$G_{(2m-b,b)} = [u^b]((1-u)(1+su+u^2)^m). \tag{2}$$

The whole two-row family is thus encoded in the single degree- $(2m+1)$ polynomial $(1-u)(1+su+u^2)^m \in \mathbb{Z}[i][u]$, and the conjecture asserts that none of its coefficients $u^1, \dots, u^m$ vanishes, except u^2 at $m = 2$.

2 The imaginary part is an alternating trinomial sum

Lemma 1. *Let $T(N, j) = [u^j](1 + u + u^2)^N$ for $N \geq 0$ (so $T(N, j) = 0$ unless $0 \leq j \leq 2N$, $T(N, j) = T(N, 2N - j) \geq 0$, and $j \mapsto T(N, j)$ is unimodal, strictly increasing for $0 \leq j \leq N$ when $N \geq 2$). Then for all $m \geq 1$ and $0 \leq b \leq m$,*

$$\operatorname{Im} G_{(2m-b, b)} = \sum_{k \geq 1, k \text{ odd}} (-1)^{(k-1)/2} \binom{m}{k} [T(m-k, b-k) - T(m-k, b-k-1)]. \quad (3)$$

Proof. Split the quadratic into its real and imaginary parts as a polynomial in the *real* indeterminate u :

$$1 + su + u^2 = (1 + u + u^2) + iu.$$

By the binomial theorem,

$$(1 + su + u^2)^m = \sum_{k=0}^m \binom{m}{k} (iu)^k (1 + u + u^2)^{m-k}.$$

The coefficients of $(1 + u + u^2)^{m-k}$ are real, and i^k is purely imaginary exactly for k odd, with $i^k = i(-1)^{(k-1)/2}$ there. Hence the imaginary part of the polynomial $(1 + su + u^2)^m$ (coefficient by coefficient) is

$$\operatorname{Im}[(1 + su + u^2)^m] = \sum_{k \text{ odd}} (-1)^{(k-1)/2} \binom{m}{k} u^k (1 + u + u^2)^{m-k}. \quad (4)$$

Because $1 - u$ has real coefficients, $\operatorname{Im}[(1 - u)(1 + su + u^2)^m] = (1 - u) \operatorname{Im}[(1 + su + u^2)^m]$, and extracting $[u^b]$ from $(1 - u)u^k(1 + u + u^2)^{m-k} = (u^k - u^{k+1})(1 + u + u^2)^{m-k}$ gives $T(m-k, b-k) - T(m-k, b-k-1)$. Combining with (??) proves (??). $\square$

Write

$$\Delta T(N, j) := T(N, j) - T(N, j-1), \quad c_k := \binom{m}{k} \Delta T(m-k, b-k), \quad (5)$$

so that $\operatorname{Im} G_{(2m-b, b)} = \sum_{k \text{ odd}} (-1)^{(k-1)/2} c_k = c_1 - c_3 + c_5 - \dots$.

Lemma 2. *For every term in (??) with $1 \leq k \leq b$ one has $c_k \geq 0$; moreover $c_k = 0$ for $k > b$. Consequently $\operatorname{Im} G_{(2m-b, b)}$ is an alternating sum of finitely many non-negative integers $c_1, c_3, c_5, \dots$, the last index being $\leq b$.*

Proof. If $k > b$ then $b-k < 0$, so $T(m-k, b-k) = T(m-k, b-k-1) = 0$ and $c_k = 0$. For $1 \leq k \leq b$ set $N = m-k \geq 0$ and $j = b-k$ with $0 \leq j \leq N$ (the upper bound is $b \leq m$). Trinomial coefficients are unimodal and symmetric about $j = N$ and non-decreasing on $0 \leq j \leq N$, whence $\Delta T(N, j) = T(N, j) - T(N, j-1) \geq 0$ and $c_k \geq 0$. (For $j = 0$, $\Delta T(N, 0) = 1$.) $\square$

The reduction of the conjecture is now immediate.

Proposition 1 (Reduction). *If $\operatorname{Im} G_{(2m-b, b)} \neq 0$ then $G_{(2m-b, b)} \neq 0$. Hence the two-row $d=4$ law, $G_{(2m-b, b)} = 0 \iff (a, b) = (2, 2)$, follows from*

$$\operatorname{Im} G_{(2m-b, b)} \neq 0 \quad \text{for all } 1 \leq b \leq m \text{ with } (m, b) \neq (2, 2), \quad (6)$$

together with $\operatorname{Im} G_{(2m-2, 2)} = m(m-2)$, which vanishes (among valid shapes) only at $m = 2$.

3 The boundary families: closed forms

The smallest values of $|G|$ — hence the cases that “almost” vanish — occur at the extreme second rows. All four are closed in (??).

Theorem 1 ($b = 1, 2, 3, 4$). *For the four boundary families,*

$$\begin{aligned} b = 1 : \quad G_{(2m-1,1)} &= (m-1) + m i, & \text{Im } G &= m, \\ b = 2 : \quad G_{(2m-2,2)} &= m(m-2) i, & \text{Im } G &= m(m-2), \\ b = 3 : \quad \text{Im } G_{(2m-3,3)} &= \frac{1}{3} m(m-1)(m-2), \\ b = 4 : \quad \text{Im } G_{(2m-4,4)} &= \frac{1}{3} m(m-1)(2m-7). \end{aligned}$$

Consequently $\text{Im } G_{(2m-b,b)} \neq 0$ for $b \in \{1, 2, 3, 4\}$ on the whole valid range $b \leq m$, except $(m, b) = (2, 2)$ where $G = 0$; and $(2, 2)$ is the unique vanishing two-row fiber with second row ≤ 4 .

Proof. $b = 1$. Directly from (??): $r_0 = 1$ and $r_1 = [u^1](1 + su + u^2)^m = ms = m(1 + i)$, so $G_{(2m-1,1)} = r_1 - r_0 = (m-1) + m i$. Thus $\text{Im } G = m \geq 1$.

$b = 2$. In (??) only $k = 1$ contributes ($k = 3 > b$ gives $c_3 = 0$): $\text{Im } G = m \Delta T(m-1, 1) = m(T(m-1, 1) - T(m-1, 0)) = m((m-1) - 1) = m(m-2)$. For the real part, k even contributes nothing to Im , and a parallel split of the real part (or the direct computation $r_2 = m + m(m-1)i$, $r_1 = m + m i$) gives $\text{Re } G_{(2m-2,2)} = 0$. Hence $G_{(2m-2,2)} = m(m-2) i$, which vanishes for a valid shape ($m \geq 2$) only at $m = 2$, i.e. at $(2, 2)$.

$b = 3$. Terms $k = 1, 3$. Using $T(N, 0) = 1$, $T(N, 1) = N$, $T(N, 2) = \binom{N+1}{2}$:

$$c_1 = m \Delta T(m-1, 2) = m \left(\binom{m}{2} - (m-1) \right) = \frac{1}{2} m(m-1)(m-2), \quad c_3 = \binom{m}{3} \Delta T(m-3, 0) = \binom{m}{3} = \frac{1}{6} m(m-1)(m-2).$$

So $\text{Im } G = c_1 - c_3 = \left(\frac{1}{2} - \frac{1}{6} \right) m(m-1)(m-2) = \frac{1}{3} m(m-1)(m-2)$, nonzero for integer $m \geq 3$.

$b = 4$. Terms $k = 1, 3$. Counting step-sequences in $\{0, 1, 2\}^N$ by their sum gives $T(N, 2) = \binom{N}{2} + N$ and $T(N, 3) = \binom{N}{3} + N(N-1)$, hence $\Delta T(N, 3) = \binom{N}{3} - \binom{N}{2} + N(N-3)$. With $N = m-1$,

$$c_1 = m \Delta T(m-1, 3) = m \left[\binom{m-1}{3} - \binom{m-1}{2} + (m-1)(m-4) \right], \quad c_3 = \binom{m}{3} \Delta T(m-3, 1) = \binom{m}{3} (m-4),$$

using $\Delta T(m-3, 1) = T(m-3, 1) - T(m-3, 0) = (m-3) - 1 = m-4$. Subtracting and simplifying,

$$\text{Im } G_{(2m-4,4)} = c_1 - c_3 = \frac{1}{3} m(m-1)(2m-7)$$

(verified as a polynomial identity). For integer $m \geq 4$ the factor $2m-7$ is odd, hence nonzero, so $\text{Im } G \neq 0$. $\square$

Remark 1 (Robustness: the margin is $\geq m$, not $\frac{1}{4}$). The hook $b = 1$ has $|G_{(2m-1,1)}|^2 = m^2 + (m-1)^2$, an *increasing* quantity. A prior reduction passed to recessive continued-fraction coordinates ξ_n and a normalised positivity $P_n = q_n/n$ whose margin tended to $\frac{1}{4}$; that knife-edge is an artifact of dividing by n . In the native Gaussian-integer coordinate the binding gap grows linearly in m (Theorem ??), so the law is not marginal at all.

4 The hook is the global minimiser

Theorem 2 (Minimum over the second row). *For every $m \geq 4$,*

$$\min_{1 \leq b \leq m} |\text{Im } G_{(2m-b,b)}| = m,$$

attained (uniquely) at the hook $b = 1$; for $m = 3$ the minimum is 2, attained at the rectangle $b = 3$. In particular $|\operatorname{Im} G_{(2m-b,b)}| \geq 1$ for all $m \geq 3$, so $G_{(2m-b,b)} \neq 0$, for every $1 \leq b \leq m$.

This is verified exhaustively for $3 \leq m \leq 300$ and all $1 \leq b \leq m$ (§??). The cases $m \leq 2$ are finite: $m = 1$ gives only $G_{(1,1)} = i$; $m = 2$ gives $G_{(3,1)} = 1 + 2i$ and $G_{(2,2)} = 0$. Theorem ?? is therefore equivalent to the conjecture, and a proof of it (for all m) closes the two-row law. What remains open is exactly the uniform non-vanishing (??) for $b \geq 5$, established above only for $b \leq 4$.

5 Where the residual gap lives, and two routes

The integer roots are all out of range. View $\operatorname{Im} G_{(2m-b,b)}$, for fixed b , as a polynomial $I_b(m) \in [m]$ of degree $2b$ (its leading term matches that of $|G|^2$ up to the obvious square structure). Computation of I_b for $b \leq 15$ shows that its *integer* roots are exactly $m \in \{0, 1, \dots, \lfloor (b-1)/2 \rfloor\}$ — all strictly less than b , hence outside the valid range $m \geq b$ — while every root of I_b in $[b, \infty)$ is irrational. Thus the law for the family b is equivalent to:

$$I_b(m) \text{ has no integer root } \geq b \quad (\text{the single exception } I_2(2) = 0 \text{ is the } (2,2) \text{ fiber}). \quad (7)$$

The integer roots $0, \dots, \lfloor (b-1)/2 \rfloor$ are forced linear factors $m, (m-1), \dots$ of I_b (degenerate shapes); the content of (??) is that the complementary “tail” factor has no integer zero. This is a genuine Diophantine statement per b ; the difficulty is to make it uniform in b .

Route A (2-adic / level-2 congruence). Both $\binom{m}{k}$ and $T(N, j) \bmod 2$ are governed by base-2 digit rules (Lucas; the self-similar trinomial pattern). The leading 2-adic term of the alternating sum (??) should be computable and nonzero, yielding a finite valuation $v_2(\operatorname{Im} G)$ and hence non-vanishing. This is the two-row instance of the “level-2 congruence π^2 -digit = $\text{bit}_1(f^\lambda) \oplus \text{bit}_1(\chi^\lambda(2^m))$ ” lever recorded in the 4-core valuation programme.

Route B (ballot / sign of an alternating unimodal sum). By Lemma ??, $\operatorname{Im} G = c_1 - c_3 + c_5 - \dots$ with $c_k \geq 0$. Each $\Delta T(N, j)$ is itself a ballot-type count ($\Delta T(N, j) = [u^j] (1-u)(1+u+u^2)^N$, a difference of trinomial heights). A reflection/ballot model exhibiting $\operatorname{Im} G$ as a *signed* lattice-path count with a sign-reversing involution whose fixed points are nonempty would give $\operatorname{Im} G \neq 0$ directly, and is the natural candidate to also recover the sharp bound $|\operatorname{Im} G| \geq m$ of Theorem ??.

6 Computational verification

All identities were checked in two independent ways: (i) by expanding $(1 + su + u^2)^m$ symbolically over $\mathbb{Z}[i]$ and reading off $r_b - r_{b-1}$, and (ii) by the trinomial alternating sum (??) evaluated in exact integer arithmetic (cached trinomial rows). The two agree for all tested (m, b) . The exhaustive search over $3 \leq m \leq 300$, $1 \leq b \leq m$, confirms: $\operatorname{Im} G_{(2m-b,b)} \neq 0$ except at $(m, b) = (2, 2)$; the global minimum $\min_b |\operatorname{Im} G| = m$ for $m \geq 4$ at $b = 1$ (and $= 2$ at $m = 3$, $b = 3$); and the closed forms of Theorem ??. Scripts: `~/projects/scratch/2026-06-06-prove/(rl.py, closed.py, imstruct.py, improfile2.py, verify.big.py)`.